These characteristics are the motivations for a decentralized application that independently verifies that students have fulfilled degree or certificate requirements. This application is a self-help tool for key stakeholders in a degree or certification pathway to monitor progress through a program.

The use case in this chapter is a real certificate program at the University at Buffalo. This certificate program is for data-intensive computing; the detailed requirements are at http://mng.bz/4Bja. I am using this scenario to motivate you to look for issues in your environment that can be addressed by blockchain solutions and can benefit from the blockchain-based approach. That's an essential goal of the problem discussed in this chapter.

11.2 The roadmap

Chapters 2-10 covered pieces of the puzzle in the development of a blockchain-based Dapp. In this chapter, all these pieces are assembled to solve a single problem. It is a challenge to view the concepts together, so here is a roadmap to guide you through the chapter and the Dapp development process. Figure 11.1 closely maps the sections of this chapter, each section demonstrating a task in the roadmap. Review this roadmap before starting the next section.